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Project Link: <https://www.programiz.com/online-compiler/55rHLx53GVtpW>

-This should open a website with the code, you all can work on your chromebooks for this or on a shared computer. If it does not load correctly try reopening it.

Story: Alice is very hungry. Every day she yearns for food but lacks a physical body due to her digital form, and thus cannot get it. Because of this she is on the verge of starvation. To solve this problem, your task is to create a kitchen using object oriented programming which she can interact with to satisfy her everlasting desire.

Things to know:

- We will be using the Object class to represent both people, and objects in the scenario
- We will be using the kitchen class to represent the physical space the objects are in
- You can reuse previously created methods when creating new methods
- The project should get harder as you go, but isn't meant to be impossible, so if that is the case feel free to ask for help

Steps:

1. Go to the Object class and finish the implementation of the method "Public Object(String name) {}" (this is the method that gets called when an instance of the Object class is created) so that it changes the objects name to whatever name the user wants
2. Create an instance of the Object class named "Alice"
3. Finish the implementation of the method getName() which returns the name of the person.
4. In main print out the users name

Note: the following steps may be challenging but give it your best effort and don't be afraid to ask for help

5. In Kitchen, finish the placeObject method so that when called and inputting an object, that object gets placed in the corresponding row/column (hint: how do you traverse a list/array?)
6. Place the Object Alice in the second square of the third row (position (3, 2)) then call printKitchen in main to see if it worked
7. Finish the moveObject method so when called and given a direction ("up", "down", "left", "right") it moves the object in that direction in the kitchen (hint: you may need a way to get the objects current position)
8. In Main make a sequence of events to test this by printing calling the printKitchen method and the methods you have now created
9. The kitchen is quite small at the moment, make it a 5 by 7 kitchen by changing the necessary arrays and update the printKitchen method to account for this

BONUS CHALLENGE: Make a method where you can choose the dimensions of the kitchen. You will need to make it so the for loops in all of your methods involving the arrays don't use a constant number (like how it was originally until i and k reached 3) to determine when to stop looping.

10. Now with your 5x7 kitchen create the following objects and place them wherever you'd like: Fridge, Sink, Stove

Note: from now on you will need to determine how to approach implementations from scratch (no guidance on methods and variables). Instructions will be intentionally vague to hopefully foster problem solving and creative idea generation. There is no limit to how you can solve the problems unless the idea explicitly breaks something.

11. Complete the Use Object method so that Alice can use things like the fridge and sink (must be realistic as in cannot just be useable from anywhere)
12. Create an inventory for Alice to store objects
13. Create a way to store fruits and vegetables of your choice in the fridge
14. Make it so Alice can grab fruits and vegetables
15. Make it so Alice can wash the fruits and vegetables in the sink
16. Make it so Alice can eat the ones in her inventory, but only if they are clean
17. Find a way to place pots on the stove
18. Make it so Alice can create a meal of your choice using the fruits and vegetables (must follow a recipe of your choice, ex must include tomato, cucumber, and grapes to create the meal called "atrocious")
 - a. The ingredients used should be removed from inventory and replaced by the meal
 - b. Meal should behave the same way as the other foods (ex can be eaten and used in recipes)
 - c. **BONUS:** Make additional recipes using meals as ingredients
19. Make a health bar for Alice that gets filled when eating
20. String all of your methods together in main to finish the story and have Alice collect food, make a meal, and eat the meal.

THE END

The project is now done. Feel free to add whatever you'd like to the file and have fun.

What you practiced:

- Creating instances of classes (very important in java)
- Creating methods that interact with other classes
- Abstraction (the art of ignoring things that have already been implemented and just using their methods without caring about how they work)
- Critical thinking
- How to use basic data structures like arrays
- Making/changing variables in objects/classes
- Object Oriented Programming (this is usually its own separate college course and is very important)
- Likely more than just this

What next:

Now that you are hopefully familiar with using Java, we can move on to learning how to program FRC robots like the ones that we use in competitions using a library called WPILib.